

Random Variable - Practice Problems

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1. Random Variable Basics (Page 1-3)

Problem 1.1

Tossing a fair coin twice. x is a random variable representing the number of heads appearing when two coins are tossed simultaneously.

Find:

$$P(x = 0), \quad P(x = 1), \quad P(x = 2), \quad P(|x| \leq 1)$$

Problem 1.2

Y is a random variable that represents the number of heads appearing when three coins are tossed simultaneously.

Find:

$$P(Y \leq 3)$$

Also find:

$$P(Y^2 + 2 \leq 6)$$

Problem 1.3

You roll a fair dice. $S = \{1, 2, 3, 4, 5, 6\}$. Random variable X : (number appearing on the dice)² - 6.

Find:

$$P\left[\left(\frac{X^2 + 4}{2} - 10\right) > 0\right]$$

Problem 1.4

A group of students typically take 30 min to 60 min to solve a Quiz. A random variable X is defined such that it represents the time taken by a randomly selected student to solve the Quiz.

- (a) What all values can X take?
- (b) Find $P(X = 35)$ and $P(X = 43.5)$.

Problem 1.5

Tossing 3 coins simultaneously. x : number of heads appearing. Draw the PMF of R.V. x .

Problem 1.6

x : number of heads appearing when 3 coins are tossed simultaneously. Draw the CDF of R.V. x .

Find:

$$F_x(0.5), \quad F_x(1), \quad F_x(-1), \quad F_x(1.5), \quad F_x(-0.5), \quad F_x(0), \quad F_x(3)$$

2. CDF and Probability Calculations (Page 4-6)

Problem 2.1

CDF of R.V. x is given. Find:

- (a) $P(-1 < x \leq 4)$
- (b) $P(-1 < x < 4)$
- (c) $P(-1 \leq x < 4)$
- (d) $P(-1 \leq x \leq 4)$
- (e) $P(x = 2)$
- (f) $P(x = 4)$

Problem 2.2

Find which of the following CDF is valid. (Check properties: non-decreasing, $F_x(-\infty) = 0$, $F_x(\infty) = 1$, $F_x(a^+) = F_x(a)$)

Problem 2.3

CDF of a R.V. X is shown. X : Mixed R.V.

Find:

$$P(X = 1), \quad P(X = 2), \quad P(X = -2), \quad P(-1 < X \leq 3), \quad P(1 \leq X < 4)$$

3. PDF, Mixed R.V., and Valid PDF (Page 6-9)

Problem 3.1

x : number of heads appearing when 3 coins are tossed simultaneously. Draw PMF, PDF & CDF of R.V. x .

Problem 3.2

x : Mixed R.V. PDF of R.V. x is shown.

- (a) Find α .
- (b) Find $P(x = 1)$.
- (c) Find $P(x = 3/2)$.
- (d) Find $P(1 \leq x < 3/2)$.

Problem 3.3

Valid PDF = ? (Identify which of the given functions can be a valid PDF)

Problem 3.4

PDF of RV x is given:

$$f_x(x) = ke^{-3|x|}; \quad -\infty < x < \infty$$

Find the value of k .

Problem 3.5

PDF of RV x is shown. Find the CDF.

Problem 3.6

PDF: $f_x(x) = ae^{-b|x|}$; $b > 0$, $a > 0$. Find CDF for $x > 0$ and for $x \leq 0$.

Problem 3.7

Find the following probabilities in terms of CDF and PDF:

- (a) $P(x < 2)$
- (b) $P(x > 2)$
- (c) $P(x \geq 2)$
- (d) $P(|x| > 2)$
- (e) $P(|x| \leq 2)$

4. Symmetry of R.V. (Page 10)

Problem 4.1

A R.V. x is called symmetrical about 0 if $\forall x \in \mathbb{R}$:

$$P[X \geq x] = P[X \leq -x]$$

$F_x(x)$ is CDF of x where x is symmetrical about 0.

Find:

(i) $P\{|x| \leq x\}$ is:

(a) $1 - 2F_x(x)$ \quad (b) $1 + 2F_x(x)$ \quad (c) $2F_x(x) - 1$ \quad (d) $F_x(x) - 1$

(ii) $P\{|x| > x\}$ is:

(a) $2[1 - F_x(x)]$ \quad (b) $2[1 + F_x(x)]$ \quad (c) $1 - F_x(x)$ \quad (d) $1 + F_x(x)$

(iii) $P\{x = x\}$ is:

(a) $F_x(x) - F_x(-x) - 1$ \quad (b) $F_x(x) + F_x(-x) + 1$

(c) $F_x(x) - F_x(-x) + 1$ \quad (d) $F_x(x) + F_x(-x) - 1$

5. Conditional Probability & Bayes' Theorem (Page 11-12)

Problem 5.1

A die is rolled. What is the probability that the number is even, given that it is greater than 2?

Problem 5.2

Suppose:

- 1% of people have a disease: $P(D) = 0.01$, $P(\text{No } D) = 0.99$
- Test is 99% accurate:
 - $P(\text{Positive} \mid D) = 0.99$
 - $P(\text{Positive} \mid \text{No } D) = 0.01$

What is the probability that a person actually has the disease given their test is positive?

Problem 5.3

CDF of RV Y is given as:

$$F_Y(y) = (1 - e^{-y})U(y)$$

Find:

$$P\left[\frac{Y < 2}{Y > -1}\right]$$

Problem 5.4

Let sample space $S = \{s_i\}_{i=1}^4$.

$$P(s_1) = \frac{1}{6}, \quad P(s_2) = \frac{1}{3}, \quad P(s_3) = \frac{1}{3}, \quad P(s_4) = \frac{1}{6}$$

R.V. $X(s_i)$ is defined such that $X(s_i) = i - 2$; $1 \leq i \leq 4$. Draw PMF.

6. Double Integrals over Regions (Page 13-14)

Problem 6.1

Draw region R : $x > 0, y > 0, x + y \leq 2$.

Find:

$$\iint_R dx \cdot dy, \quad \iint_R xy \, dx \, dy$$

Problem 6.2

Draw region R' : $x > 0, y > 0, x + y \leq 2, x + y \geq 1$.

Find:

$$\iint_{R'} dx \cdot dy, \quad \iint_{R'} xy \, dx \, dy$$

7. Joint Distributions - 2D Random Variables (Page 15-23)

Problem 7.1

x and y are two R.V. such that $x \in \{0, 1, 2\}$ (DRV), $y \in \{1, 2, 3\}$ (DRV).

Joint PMF $P(x = x, y = y)$:

	y=1	y=2	y=3
x=0	0.15	0.05	0.1
x=1	0.1	0.03	0.17
x=2	0.05	0.25	0.1

Find:

$$P(0 < x \leq 2, 1 < y \leq 3)$$

Also find $P(y = 2)$.

Problem 7.2

$$f_{XY}(x, y) = \begin{cases} be^{-(x+y)} & ; 0 < x < 2 \text{ and } 0 < y < \infty \\ 0 & ; \text{otherwise} \end{cases}$$

(i) Find $b = ?$

(ii) $P(X + Y < 1) = ?$

(iii) $P(X + Y = 1) = ?$

Problem 7.3

Joint CDF of x and y :

$$F_{xy}(x, y) = \begin{cases} \frac{5}{4} \left[\frac{x+e^{-(x+1)y^2}}{x+1} - e^{-y^2} \right]; & 0 \leq x \leq 4, y \geq 0 \\ 0 & ; x < 0 \text{ and } y < 0 \\ 1 + \frac{1}{4}e^{-5y^2} - \frac{5}{4}e^{-y^2} & ; x \geq 4 \text{ and } y \geq 0 \end{cases}$$

Find:

$$P[2 < x \leq 3], \quad P[2 < x \leq 3, 1 < y \leq 4]$$

Problem 7.4

The joint PDF of X and Y is given by:

$$f_{xy}(x, y) = xye^{-(x^2+y^2)/2}u(x)u(y)$$

- (a) Find the marginal PDFs $f_x(x)$ and $f_y(y)$, and the conditional PDFs $f_{x|y}(x|y)$ and $f_{y|x}(y|x)$.
 (b) Are X and Y independent?

Problem 7.5

Let the joint probability density function of random variables X and Y be given by:

$$f_{X,Y}(x, y) = \begin{cases} kxy & 0 < x < 1, 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

1. Find the value of K .
2. Compute the conditional probability:

$$P\left(y < \frac{1}{8} \mid x < \frac{1}{2}\right)$$

Problem 7.6

$$f_{xy}(x, y) = cx(x - y); \quad 0 < x < 2; \quad -x < y < x$$

Find:

$$P[0 < Y < 2] = ?$$

Problem 7.7

Consider the joint PDF of two random variables X and Y :

$$f_{X,Y}(x, y) = \begin{cases} 8xy, & 0 \leq x \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Determine the conditional PDF:

$$f_{Y|X}(y | x = \frac{1}{2})$$

Problem 7.8

The joint PMF is given as:

$$P(x, y) = K(2x + 3y); \quad x \in \{0, 1, 2\}, y \in \{1, 2, 3\}$$

Find:

(i) $P(y = 1), P(y = 2), P(y = 3), P(x = 0), P(x = 1), P(x = 2)$

(ii) $P\left(\frac{x \geq 1}{y \leq 1}\right), P(x + y > 3)$

Problem 7.9

Joint PDF of R.V. X and Y is given as:

$$f_{xy}(x, y) = \begin{cases} 3xy; & x > 0, y > 0, x + y \leq 2 \\ 0; & \text{otherwise} \end{cases}$$

Find:

$$P[x + y \geq 1] = ?$$

8. Expectation, Mean, Variance, Moments (Page 24-28)

Problem 8.1

A random variable X represents the number of chocolates a child gets when spinning a wheel. Find $E[X]$, given:

- $P(X = 1) = 0.2$
- $P(X = 2) = 0.5$
- $P(X = 3) = 0.3$

Problem 8.2

Let X be a continuous random variable with PDF:

$$f_X(x) = 2x \quad \text{for } x \in [0, 1], \quad f_X(x) = 0 \text{ otherwise}$$

Find $E[X]$.

Problem 8.3

A random variable Y is defined as $Y = aX + b$, where a and b are constants. Find $E[Y]$ given that X is a R.V. whose PDF is $f_X(x)$.

Problem 8.4

A R.V. x is given. Another R.V. y is defined as $y = \sin x$. Find the mean of random variable y . PDF of R.V. x is given as $f_x(x)$.

Problem 8.5

The PDF of R.V. is shown (uniform from -1 to 3). Find its mean and variance.

Problem 8.6

A R.V. Y is defined as $Y = aX + b$. Find mean and variance of Y in terms of mean and variance of X .

Problem 8.7

$Y = -2X + 3$, $\mu_X = 3$, $\sigma_X^2 = 4$. Find σ_Y^2 and μ_Y .

Problem 8.8

A R.V. X takes values from 1 to 5 with probabilities as shown:

x	1	2	3	4	5
P(x=x)	0.1	0.2	0.4	0.2	0.1

Find $E[X]$, $E[X^2]$, and σ_X^2 .

Problem 8.9

PDF of a R.V. is given:

$$f_x(x) = \frac{1}{2}|x|e^{-|x|}; \quad -\infty < x < \infty$$

Find mean, variance and 11th order moment about origin.

Problem 8.10

Let the R.V. x represent the number of times a dice needs to be thrown till 3 is observed for the first time. Find expectation of x .

9. Joint Moments, Covariance, Correlation (Page 29-34)

Problem 9.1

$w = ax + by$. Find mean, MSV and variance of w .

Problem 9.2

$w = 3x - 2y$. Given $\bar{x} = 1/2$, $\bar{y} = 2$, $\overline{x^2} = 5/2$, $\overline{y^2} = 19/2$, $C_{xy} = -\frac{1}{2\sqrt{3}}$.

- (i) Find mean, MSV and variance of w .
- (ii) Find correlation coefficient of R.V. x and y .

Problem 9.3

$$f_{xy}(x, y) = \begin{cases} \frac{xy}{9}; & 0 < x < 2 \text{ and } 0 < y < 3 \\ 0; & \text{otherwise} \end{cases}$$

Find covariance of x and y . Check independency, orthogonality and un-correlation.

Problem 9.4

$z = -8x + 4y$. Find variance of R.V. z when:

- (a) R.V. x and y are uncorrelated.
- (b) R.V. x and y are orthogonal.
- (c) R.V. x and y are independent.

Problem 9.5

DRV $x \in \{-1, 0, 1\}$; $P(x = -1) = P(x = 0) = P(x = 1) = 1/3$. $y = x^2$.

Find $E[x]$, $E[y]$, $E[xy]$ =?

Comment on independency, orthogonality and uncorrelation.

Problem 9.6

If $f_x(x) = f_x(-x)$ (PDF of R.V. x is even), find $E[x]$.

Problem 9.7

$$f_{xy}(x, y) = \begin{cases} \frac{6}{\pi}(x^2 + y^2); & x^2 + y^2 \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

Comment on orthogonality, independency and un-correlation.

Problem 9.8

$$f_{xy}(x, y) = \begin{cases} 6(1 - y); & 0 \leq x \leq y \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

Are x and y independent?

Problem 9.9

x and y are two R.V.; $\bar{x} = 1$, $\bar{y} = 2$, $\sigma_x^2 = 6$, $\sigma_y^2 = 9$. $\rho_{xy} = -2/3$.

Find covariance and correlation.

Problem 9.10

Correct statement?

- (a) If x and y are orthogonal then $\sigma_{xy} = -\mu_x\mu_y$.
 - (b) If x and y are uncorrelated then $R_{xy} = \mu_x\mu_y$.
 - (c) If x and y are orthogonal then $R_{xy} = 0$.
 - (d) If x and y are uncorrelated then $\sigma_{xy} = 0$.
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10. Q-Function and Error Function (Page 34-35)

Problem 10.1

Your answer came out to be $Q(0.4)$ but all the options are mentioned in terms of $P(x)$ where:

$$P(x) = \int_0^x e^{-t^2} dt$$

Write $Q(0.4)$ in terms of $P(x)$.

Options:

- (a) $\frac{1}{2} - \frac{1}{\sqrt{\pi}} P\left(\frac{0.4}{\sqrt{2}}\right)$
 - (b) $\frac{1}{2} + \frac{1}{\sqrt{\pi}} P\left(\frac{0.4}{\sqrt{2}}\right)$
 - (c) $\frac{1}{2} - \frac{1}{\sqrt{\pi}} P(0.2)$
 - (d) $\frac{1}{2} + \frac{1}{\sqrt{\pi}} P(0.2)$
-

11. Standard Distributions: Uniform (Page 35-36)

Problem 11.1

$x \sim U[-1, 3]$; find mean, MSV and variance.

Problem 11.2

A R.V. $x = 3 \cos \theta$, where R.V. $\theta \sim U[0, 2\pi]$. Find variance of R.V. x .

12. Standard Distributions: Gaussian / Normal (Page 37-40)

Problem 12.1

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Find:

$$\int_{-\infty}^{\infty} e^{-(x-2)^2} dx = ?$$

Problem 12.2

$$I = \int_{-\infty}^{\infty} x^2 \exp \left[-\frac{(x-2)^2}{4} \right] dx$$

Find $I = ?$ **Problem 12.3**

$$I = \int_0^{\infty} x e^{-x^2} dx$$

Problem 12.4

$$f_x(x) = k e^{-x^2/2}$$

Find $k = ?$ **Problem 12.5**

$$I = \int_2^{\infty} e^{-(x-2)^2/30} dx$$

Problem 12.6

$$x \sim N(\mu_x, \sigma_x^2)$$

Find $P(x > a) = ?$ Express in terms of Q -function.**Problem 12.7**

$$x \sim N(\mu_x, \sigma_x^2)$$

Find $P(a < x \leq b)$, $P(a \leq x < b)$, $P(a \leq x \leq b)$ in terms of Q -function.**13. Other Distributions (Page 41-42)****Problem 13.1** $x \sim \Delta(-1, 2, 4)$ (Triangular distribution).

$$f_x(x) = \begin{cases} \frac{2}{15}(x+1); & -1 \leq x \leq 2 \\ -\frac{x}{5} + \frac{4}{5}; & 2 \leq x \leq 4 \\ 0; & \text{otherwise} \end{cases}$$

- (i) Find mean of x .
- (ii) $P(0 < x \leq 2) = ?$
- (iii) $E[x] = ?$

Problem 13.2

$$f_x(x) = \begin{cases} \frac{x}{3} e^{-x^2/6} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

Find mean.

Problem 13.3

$$I = \int_{-\infty}^{\infty} (x+3)^2 e^{-3(x+3)^2} dx = ?$$

Problem 13.4

$$f_X(x) = \frac{1}{\sqrt{\pi}} e^{-(x-0.5)^2}$$

Given that $P(X > L) = 0.16$.

Find:

- (a) $P(0.5 < X < L)$
- (b) $P(X < 0.5)$
- (c) $P(X \leq 0)$
- (d) $F_X(L)$
- (e) $P(|X - 0.5| < 0.5)$

14. Linear Transformation of Random Variables (Page 43-44)

Problem 14.1

$x \sim U[-2, 3]$. $Y = 2x - 3$.

- (i) Find the PDF of Y .
- (ii) Find $E[Y]$.
- (iii) Find $P[Y > 2]$.

Problem 14.2

$x \sim N(2, 4)$. Another R.V. Y is defined by the relation $Y = -x + 3$.

- (i) Find PDF of Y .
(ii) Find $P(-1 < Y < 8)$.

Problem 14.3

$$f_x(x) = \begin{cases} x & ; 0 \leq x < 1 \\ 2 - x & ; 1 \leq x \leq 2 \\ 0 & ; \text{otherwise} \end{cases}$$

$$Y = 3x - 1.$$

- (a) Sketch $f_Y(y)$ and define it.
(b) Find $E[Y]$.
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15. DRV Transformation & Binary R.V. (Page 45-47)

Problem 15.1

Let $x \in \{0, 1\}$ represent a binary R.V. indicating the bit sent in a digital communication system.

$$P_x(0) = 0.7, \quad P_x(1) = 0.3$$

Let the modulator signal be $S = 2x - 1$. $N \sim N(0, 1)$. Received signal $R = S + N$.

- (a) Find the PMF of S .
(b) Find the expected value of R .

Problem 15.2

$x \in \{-2, -1, 0, 1, 2\}$, $P_X(x) = 1/5$ for each x .
R.V. Y is defined as $Y = x^2$.

- (a) Find the PMF of Y .
(b) Find variance of Y .

Problem 15.3

Let $x, z \in \{0, 1\}$ be independent DRVs with:

$$P_x(0) = 0.6, \quad P_x(1) = 0.4$$

$$P_z(0) = 0.7, \quad P_z(1) = 0.3$$

$$Y = x \oplus z \text{ (XOR)}.$$

- (a) Find PMF of Y .
(b) Find $E[Y]$.

Problem 15.4

Let $x \in \{1, 2, 3, 4\}$ with $P(x = 1) = 3/8$, $P(x = 2) = 1/4$, $P(x = 3) = 1/8$, $P(x = 4) = 1/4$.

$$Y = \begin{cases} x & ; \text{even} \\ 5 - x & ; \text{odd} \end{cases}$$

Find $F_Y(4) - F_Y(2)$.

Problem 15.5

Let $x_1, x_2 \in \{1, 2, 3\}$ be i.i.d. random variables with:

$$P(x) = \begin{cases} 0.2 & ; x = 1 \\ 0.5 & ; x = 2 \\ 0.3 & ; x = 3 \end{cases}$$

Define $M = \max(x_1, x_2)$ and $m = \min(x_1, x_2)$.

- (a) Find PMF of M and m .
 - (b) Are M and m independent?
 - (c) Find $P(M = 3/m = 3)$.
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16. CRV to DRV Transformation (Page 48-49)**Problem 16.1**

$x \sim U[0, 5]$. Another R.V. Y is defined by a quantization mapping (given in figure).

- (i) Find the expression of $f_Y(y)$.
- (ii) Find $E[Y]$.

Problem 16.2

A uniformly distributed random variable X with PDF:

$$f_X(x) = \frac{1}{10}(u(x + 5) - u(x - 5))$$

where $u(\cdot)$ is the unit step function, is passed through a transformation (given in figure). The PDF of the transformed random variable Y would be:

Options:

- (A) $f_Y(y) = \frac{1}{5}(u(y + 2.5) - u(y - 2.5))$
- (B) $f_Y(y) = 0.5\delta(y) + 0.5\delta(y - 1)$
- (C) $f_Y(y) = 0.25\delta(y + 2.5) + 0.25\delta(y - 2.5) + 0.5\delta(y)$
- (D) $f_Y(y) = 0.25\delta(y + 2.5) + 0.25\delta(y - 2.5) + \frac{1}{10}(u(y + 2.5) - u(y - 2.5))$

Problem 16.3

$x \sim N(0, 1)$.

$$Y = \begin{cases} 0 & ; -1 < x < 1 \\ 1 & ; x > 2 \\ -1 & ; x < -2 \\ x - 1 & ; 1 < x < 2 \\ x + 1 & ; -2 < x < -1 \end{cases}$$

Find $f_Y(y)$.

17. Order Statistics & i.i.d. R.V. (Page 50-52)**Problem 17.1**

4 R.V. x, y, z, w are i.i.d. R.V.

(a) $P(X \text{ is largest}) = ?$

(b) $P(X > Y > Z > W) = ?$

Problem 17.2

Given that $\max(x, y) > z$, which of the following statement is true?

(a) $x > z, y < z$ \quad (b) $x < z, y > z$ \quad (c) $x > z, y > z$ \quad (d) all

Problem 17.3

Given that $\max(x, y) < z$, which of the following statement is true?

(a) $x < z, y < z$ \quad (b) $x > z, y < z$ \quad (c) $x < z, y > z$ \quad (d) $x > z, y > z$

Problem 17.4

x and Y are two independent CRV.

$P(x > t_{1/2}) = 0.3, P(Y > t_{1/2}) = 0.4$.

Find $P[\max(x, Y) > t_{1/2}]$.

Problem 17.5

Given that $\min(x, y) > z$, which of the following statement is true?

(a) $x > z, y < z$ \quad (b) $x < z, y > z$ \quad (c) $x > z, y > z$ \quad (d) $x < z, y < z$

Problem 17.6

Given that $\min(x, y) < z$, which of the following statement is true?

(a) $x < z, y < z$ \quad (b) $x > z, y < z$ \quad (c) $x < z, y > z$ \quad (d) all

Problem 17.7

x and Y are two independent CRV.

$$P(x > t_{1/2}) = 0.3, P(Y > t_{1/2}) = 0.4.$$

Find $P[\min(x, Y) < t_{1/2}]$.

Problem 17.8

$x \sim U[-2, 4]$; $y \sim U[-1, 3]$; x and y are independent R.V.

$$M = \max(x, y), m = \min(x, y).$$

Find:

(a) $P(m < 0)$

(b) $P(M > 0)$

(c) $P(-1 < m < 2)$

(d) $P(-1 < M < 2)$

(e) $P(-1 < m < 0 < M < 2)$

18. Sum of Independent R.V. and CLT (Page 53-57)

Problem 18.1

$x \sim U[-2, 2]$, $y \sim U[-3, 3]$. x and y are independent R.V.

Find:

$$P[x + y \leq 2], \quad P[x + 2y - 3 < 2], \quad P[|x + y| \leq 2]$$

Problem 18.2

$x \sim N(2, 4)$; $y \sim N(3, 2)$; x and y are independent.

Find:

$$P\left(x > \frac{3y}{2}\right)$$

Problem 18.3

$x \sim N(0, 4)$; $y \sim N(0, 9)$; $\sigma_{xy} = 3$.

Find:

$$P(|x - y| > 2)$$

Problem 18.4

$x \sim N(1, 4)$; $y \sim N(2, 9)$; $z \sim N(3, 4)$; x, y, z are independent.

Find:

$$P(3x - 2y > 3z + 6)$$

Problem 18.5

x and y are two independent R.V.

$$f_x(x) = 2e^{-2x}u(x), \quad f_y(y) = 3e^{-3y}u(y)$$

Another R.V. z is defined as $z = x + y$.

(a) Find $P(z > 1)$.

(b) Find PDF of z .

Problem 18.6

u, v, w, x, y are 5 i.i.d. R.V. $\sim U[0, 1]$.

$$Z = U + 2V - W + x - 2Y$$

Find $P(Z > 1) = ?$

Options: (a) 0.5 \quad (b) 0.3 \quad (c) 0.8 \quad (d) 1

19. Non-Linear Transformation of R.V. (Page 58-59)**Problem 19.1**

$$f_x(x) = 2e^{-2x}U(x)$$

$$Y = \ln(x + 1)$$

Find PDF of Y .

Problem 19.2

$$x \sim U[0, 2]$$

$$Y = \frac{1}{x}$$

Find PDF of Y .

Problem 19.3

$$x \sim U[-1, 1]$$

$$Y = x^2$$

Find PDF of Y .

Problem 19.4

$$x \sim U[0, \pi]$$

$$Y = \cos x$$

Find PDF of Y .

Problem 19.5

$$x \sim U[-\pi, \pi]$$

$$Y = \cos x$$

Find PDF of Y .

20. Miscellaneous (Page 60-62)

Problem 20.1

U and V are i.i.d. R.V. with zero mean. CDF of U and $2V$ are $F(x)$ and $G(x)$ respectively. Correct statement for all values of x :

- (a) $F(x) - G(x) \leq 0$
- (b) $F(x) - G(x) \geq 0$
- (c) $[F(x) - G(x)] \cdot x \leq 0$
- (d) $[F(x) - G(x)] \cdot x \geq 0$

Problem 20.2

Let x_1, x_2, x_3 i.i.d. R.V. $\sim U[0, 1]$. (GATE 2014)

Find:

- (a) $P(x_1 + x_2 + x_3 \leq 1)$
- (b) $P(x_1 + x_2 + x_3 \leq 2)$
- (c) $P(x_1 + x_2 + x_3 \leq 3)$
- (d) $P(x_1 + x_2 < x_3)$